Stealth Health AI Agent – Project Submission

Candidate

Shaswat

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1. Project Overview

Stealth Health AI Agent is an AI-powered wellness coaching assistant designed to personalize health guidance using structured patient onboarding, adaptive daily check-ins, session memory, and protocol-grounded responses.

The system accepts unstructured patient onboarding information, extracts structured wellness data, adapts conversations based on the user's protocol timeline, and answers wellness questions strictly using a predefined protocol document.

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2. Live Application

Live Streamlit Application

https://stealth-health-ai-agent-qv84dd6fsmh4wrhureqd4z.streamlit.app/

Query Parameter Demo

The application supports URL-based onboarding.

Example:

https://your-app-name.streamlit.app/?raw_data=I am Guest , 22 years old. My goal is improving sleep. I currently sleep 5 hours. Put me on Day 5.

Expected Extraction Result

{
  "name": "Guest",
  "age": 22,

"primary_goal": "improving sleep",
"sleep_hours": 5,
"current_day": 5
}

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3. GitHub Repository

Repository URL:

https://github.com/shaswatgithub/Stealth-Health-AI-Agent

The repository contains:

- Source Code
- Agent Logic
- Streamlit Application
- Deployment Configuration
- Documentation
- Requirements File

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4. Problem Statement

Build a Health Coach AI Agent that:

- Accepts patient onboarding data
- Extracts structured patient information
- Performs adaptive daily check-ins
- Maintains session memory
- Answers protocol questions with minimal hallucination
- Uses a warm, clear coaching style

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5. Solution Architecture

Workflow

User Input
↓

Gemini Extraction Agent
↓
Structured Patient Profile
↓
Session Memory
↓
Day-Based Coaching Logic
↓
Protocol Grounding
↓
Personalized Response

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6. Key Features

A. Structured Patient Onboarding

The system extracts:

- Name
- Age
- Wellness Goal
- Sleep Habits
- Protocol Day

from natural language onboarding text.

Example Input:

"I am Shaswat, 22 years old. My goal is improving sleep. I currently sleep 5 hours. Put me on Day 5."

B. Adaptive Daily Check-ins

The AI dynamically changes behavior based on protocol progression.

Day 1

- Welcome message
- Goal review
- Habit baseline setup

Day 2–4

- Progress tracking
- Habit consistency monitoring

Day 5+

- Accountability coaching
- Goal-focused follow-up
- Obstacle identification

C. Session Memory

The application stores previous messages using Streamlit Session State.

This enables:

- Context retention
- Personalized conversations
- Multi-turn coaching interactions

D. Protocol-Grounded Responses

The AI is restricted to responding only from the provided wellness protocol.

If information is unavailable in the protocol, the assistant responds:

"I couldn't find that in the protocol document."

This significantly reduces hallucination risk.

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7. Technology Stack

Component	Technology
Frontend	Streamlit
AI Model	Gemini 2.5 Flash
Data Validation	Pydantic
Session Memory	Streamlit Session State

Deployment	Streamlit Community Cloud
Source Control	GitHub

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8. Agent Design

Agent State

The system maintains a structured patient profile:

```
{
  "name": "Guest",
  "age": 22,
  "primary_goal": "Improve Sleep",
  "sleep_hours": 5,
  "current_day": 5
}
```

Memory

Previous interactions are stored in session state and supplied back to the model for contextual continuity.

Reasoning

The agent combines:

- Patient Profile
- Timeline Context
- Protocol Rules
- Conversation History

to generate personalized responses.

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9. Technical Decisions

Why Streamlit?

- Fast development
- Built-in chat interface
- Easy deployment
- Excellent demo capability

Why Gemini 2.5 Flash?

- Fast inference
- Strong instruction following
- Cost-efficient
- Reliable structured extraction

Why Pydantic?

- Schema validation
- Type safety
- Clean structured system state

Why Session State?

- Lightweight memory
- No external database required
- Suitable for MVP requirements

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10. Future Improvements

Potential future enhancements include:

- PDF-based RAG pipeline
- Long-term memory storage
- User authentication
- Progress analytics dashboard
- Multi-user support
- Vector database integration
- Habit tracking visualizations

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11. Demo Video

Video Walkthrough:

[PASTE VIDEO LINK HERE]

Suggested Demo Flow:

1. Open application
2. Demonstrate onboarding extraction
3. Show parsed patient profile
4. Demonstrate Day 1 behavior
5. Demonstrate Day 5 behavior
6. Show protocol-grounded responses
7. Demonstrate session memory

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12. Conclusion

Stealth Health AI Agent successfully satisfies the MVP requirements by combining structured information extraction, adaptive coaching behavior, conversational memory, and protocol-grounded responses within a simple, deployable Streamlit application.

The solution demonstrates a practical implementation of an AI agent architecture for personalized wellness coaching while maintaining transparency and minimizing hallucinations.